The memory squeeze and your lakehouse bill

QBricks: a hedge to the cost of rising memory, Spark and compute.

Compute is being repriced by memory. Here is the hedge.

Server DRAM has become the scarcest input in the AI build-out. Every Spark billing unit carries a fixed parcel of it. This paper traces the price shock, the supply and geopolitical factors behind it, the capital chasing it — and how QBricks routes lakehouse workloads onto a Spark-free engine generation to cut the exposure by roughly 80 percent.

+93–98%
conventional DRAM contract price rise in a single quarter, Q1 2026 — a record

7×
modelled server-DRAM index at the 2027 peak vs September 2025 (index 700)

−80%
QBricks compute index at the same point: 140 vs 700 for memory-indexed Spark

Executive summary

Memory is no longer a cyclical commodity that corrects itself in eighteen months. DRAM wafer capacity is being reallocated to high-bandwidth memory for AI accelerators, hyperscalers have locked up 2027 output under multi-year agreements, and the first meaningful wave of new fabrication capacity lands only in 2027–28. Contract prices for server DRAM roughly doubled in Q1 2026 and are still rising every quarter.

This matters to data platform leaders because the billing unit of every managed Spark service — DPU, DBU, executor profile — is defined as vCPU plus a fixed block of RAM. When DRAM reprices, Spark reprices. Server makers and cloud providers began passing the increase through in late 2025.

QBricks removes most of that exposure. The lakehouse stays the governed landing zone, but roughly 70 percent of workloads are routed to a single right-sized node running a new generation of Spark-free, open-source, vectorised engines. Modelled against the same memory index, the hedged compute curve peaks at 140 rather than 700 in 2027 and stays below 115 thereafter. The commercial model is gain-share: QBricks is paid from audited savings, so the hedge funds itself.

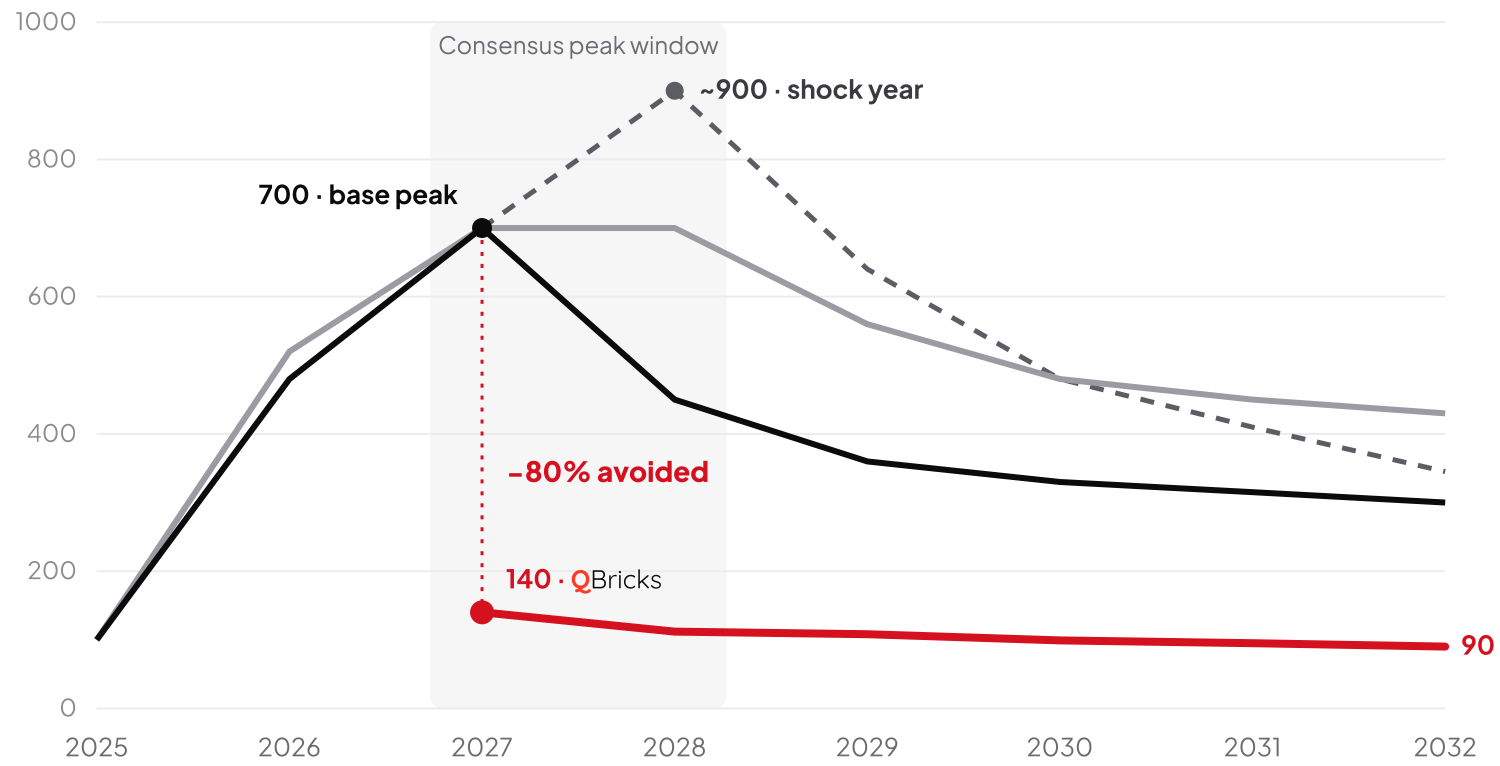
Memory cost outlook, 2026–2032

Conventional DRAM contract prices rose 45–50 percent in Q4 2025, then 93–98 percent in Q1 2026 — the largest quarterly increase on record. 64 GB RDIMMs doubled past $900. TrendForce forecasts a further 13–18 percent in Q3 2026 and continued quarterly rises through the second half of 2027, with the pace moderating only because long-term agreements now cap what the largest buyers pay; everyone else absorbs the increase. Industry reports say the three memory makers have already sold their entire 2027 DRAM and HBM output.

Our modelled scenarios below index the server DRAM contract price to September 2025 = 100. The base case (65 percent probability) peaks near 700 in 2027 and eases as new fabs ramp. An extended squeeze (35 percent) holds the plateau through 2028. A Taiwan-Strait disruption in 2028 would overlay a spike towards 900.

Server DRAM contract index vs QBricks compute index

Sep 2025 = 100 · annual average



Anchors: TrendForce contract-price data, vendor pricing. From 2026: Infinium modelled scenarios, August 2026. QBricks line: compute cost at 20% of the memory index in 2027, 25% in 2028, 30% from 2029. Forecast, not actuals.

02 · Transmission

Why this lands on Spark: the billing unit is memory

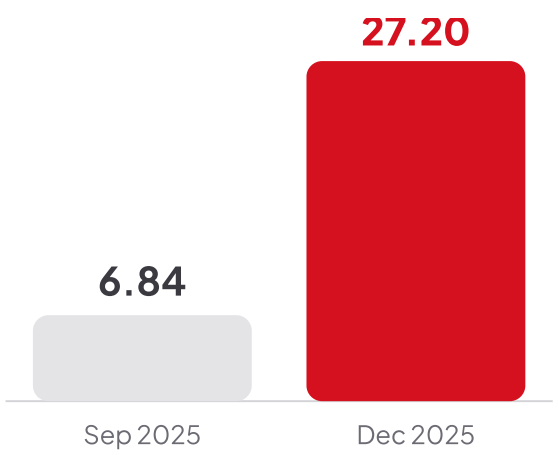
Anatomy of one Spark billing unit

1 DPU = 4 vCPU +

16 GB memory

$0.44 per DPU-hour (AWS Glue). Databricks DBUs and Dataproc profiles are built the same way: every unit of Spark compute you buy carries a fixed parcel of DRAM.

16 Gb DDR5 chip contract price, $: one DPU's 16 GB uses eight chips



+298% in three months, as DRAM capacity moved to AI accelerators (industry contract quotes).

Spark is memory-shaped by design. Executors hold partitions, shuffles and caches in RAM; memory-dense VM families are the default substrate for every managed Spark service.

So the DRAM spike is a direct input cost. The conventional DRAM that AI displaced is what fills each DPU, DBU and executor. Spark ETL sits first in line for pass-through.

Confirmed, not forecast. Server makers repriced hardware in December 2025–January 2026; OVHcloud raised cloud prices from April 2026 and again in September; memory-heavy SKUs are consistently flagged as most exposed.

03 · Supply, geopolitics and capital

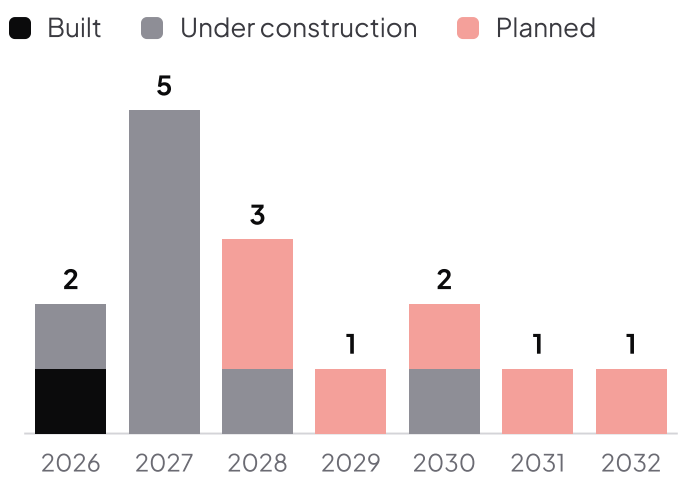
Demand runs ahead of supply for most of the window

Three forces keep the curve high. **Supply is slow.** A DRAM fab takes two to three years to build, qualify and ramp. The additions are real — SK hynix's Yongin Y1 targets 150,000 wafers a month by end-2028, Micron's Idaho and Taiwan fabs land in 2027 with Hiroshima in 2028, Samsung's new Pyeongtaek line starts in 2028 — but every HBM bit consumes three to four times the wafer area of conventional DRAM, so much of the new capacity never reaches the commodity pool.

Demand is contracted. Hyperscalers have guided to roughly $700–800 billion of 2026 capital expenditure, most of it AI infrastructure, and forecasts exceed $1 trillion a year from 2027. Each gigawatt delivered locks in years of memory consumption.

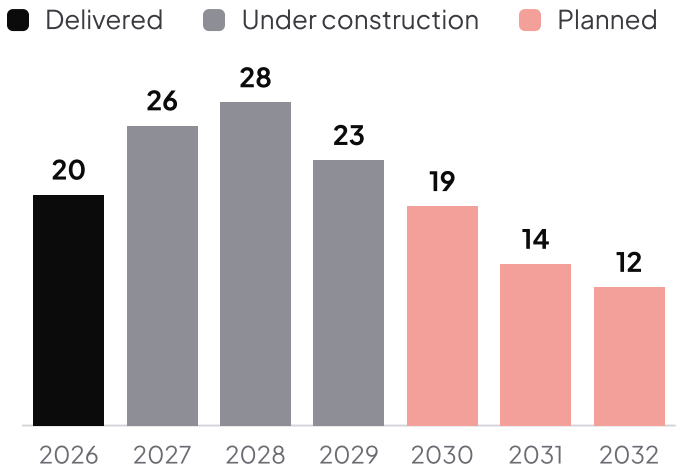
Capital is abundant. The largest hyperscalers issued about $170 billion of bonds in the first half of 2026 alone, against $121 billion for all of 2025 — money that buys memory before you do.

Supply: major fabs coming online per year



Major announced logic, DRAM and HBM fabs; public announcements, Aug 2026.

Demand: new AI data-centre capacity delivered per year, GW



AWS, Microsoft, Google, Meta, Oracle/Stargate; Infinium directional estimates from public announcements and pipeline trackers, Aug 2026.

The unhedged chain: three chokepoints, one curve

1

EUV lithography supplier worldwide — ASML (NL). Every fab on the supply chart depends on its tools.

~90%

of the most advanced logic (below 10 nm) is made in Taiwan. Maritime risk, the Strait included, has lifted chip logistics costs 15–22% in 2026.

~2/3

of DRAM is made in South Korea — geographically correlated with Taiwan. Chinese CXMT adds volume, but not HBM at scale before 2028.

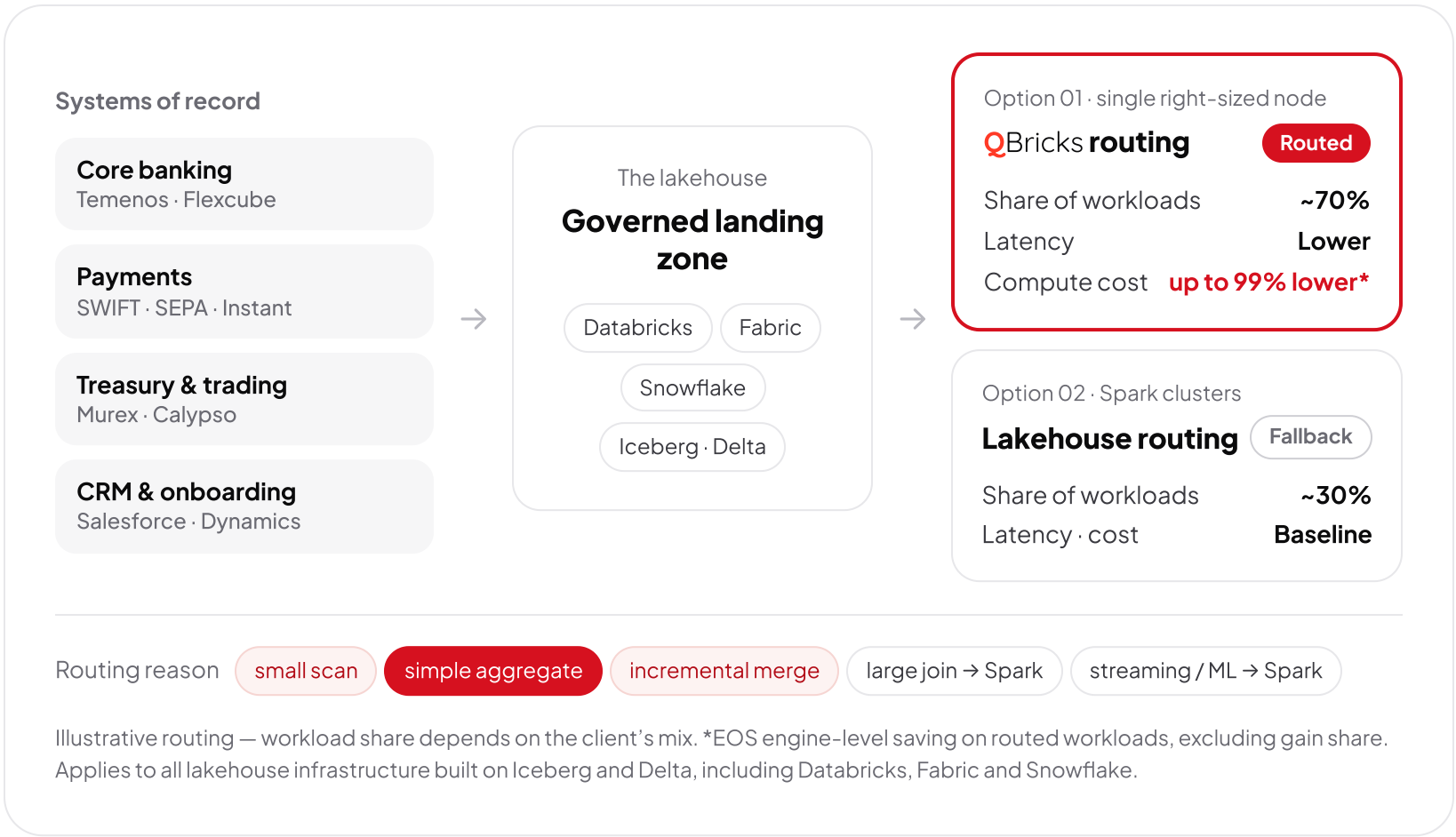
ASML EUV tools → Taiwan logic · Korea DRAM → your memory curve. Diversification (Arizona, Indiana, Idaho, Hiroshima) does not close the gap before 2028–29.

04 · The hedge

Data lands in the lakehouse. QBricks runs the workload.

The exposure comes from the shape of the engine, not from the data. Spark was designed for a world of cheap memory and expensive coordination: it parallelises across many executors, each holding a slice of the dataset in RAM, and it is billed per executor-hour. EOS, the SQL engine that powers QBricks, inverts that trade. A hardened fork of Apache DataFusion with Arrow-native streaming and the Vortex columnar format, it runs an organisation's entire pipeline estate on one right-sized VM — no Spark, low compute.

QBricks puts EOS behind its governed ingestion layer. Your lakehouse — Databricks, Microsoft Fabric, Snowflake or Oracle — remains the governed landing zone on Iceberg or Delta tables. QBricks routes each workload to EOS on a single right-sized node and returns governed data products to the landing zone for downstream consumption, or to data scientists through the Python SDK. Every record is still checked against the Open Data Contract Standard, with a human in the loop and before-and-after files for audit.



Meet EOS. The engine behind QBricks

Spark splits your data up, runs it in parallel and recompiles it — and you pay for all of that coordination before a single row is processed. EOS removes the coordination layer. Shuffle, serialisation, replication, idle autoscale headroom, per-step minimums and cross-service data movement disappear; only the work itself remains, on one committed VM.

50.6s

to ingest each TB of raw CSV as governed, query-ready data: 12.08 TB and 86.6 bn records in one run, at €0.48 compute per TB

14.5s

866 m records through the full 22-query TPC-H suite on one VM — no cluster, no tuning, no network shuffle to hide behind

1 VM

32 vCPU / 64 GiB, ≈ $31 a day, replaces the pipeline cluster. −85% compute per migrated workload; −90% cost vs managed Spark

Why EOS is different	What it means for the memory bill
Forked, not wrapped	A hardened fork of Apache DataFusion, a proven vectorised query core, tuned for the single-node execution path that managed Spark cannot take.
Arrow-native streaming	Data moves through the engine as Arrow end to end, with no serialisation between stages. Incremental changes are streamed as they land.
Vortex columnar format	State-of-the-art compression lets the engine filter and compute directly on compressed data — less RAM per row, and a smaller lakehouse storage bill.

Built with Rust	Compiled, memory-safe, no garbage collection. No JVM and no pauses: predictable low latency at a fraction of the memory footprint.
Capped compute cost	The whole pipeline estate runs on one right-sized VM as a single committed line item — forecastable to the euro, insulated from memory prices.
Governed by design	Every record is checked against an Open Data Contract Standard contract. Fully auditable, with a human in the loop.
Python SDK	Data science teams pull governed data products in a few lines of Python, in seconds — no pipeline build, no wait on engineering.

Benchmarks: ingestion of 12.08 TB TPC-H-derived CSV / 86.6 bn records in 610.8 s on nine Azure D64ds_v5 workers: 50.6 s and €0.48 timed compute per TB (29 Sep 2026, single run); TPC-H SF100 (~866 m rows), 22 queries, single Azure D32als_v6 (32 vCPU / 64 GiB), pay-as-you-go ≈ $31/day. Compute reduction is a QBricks engineering estimate of up to 85%. Cost savings are subject to the QBricks gain share. Results available on request.

05 · Revised compute cost

Worked example: the upside case

Take a bank running 20 source systems into a Databricks lakehouse, with Spark compute indexed to 100 in September 2025. This example takes the upside case: 90 percent of its workloads routed to the QBricks engine — in line with initial tests on workloads — against roughly 70 percent in a typical estate. The remainder stays on Spark clusters. The actual figure comes out of the routing assessment. Against the extended-squeeze memory index, the blended compute index becomes:

Year	Unhedged: all Spark (ext. squeeze)	Routed to QBricks (90%, upside)	QBricks cost (share)	Still on Spark	Blended	Saving
2027	700	630	**126** (20%)	70	**196**	**–72%**
2028	700	630	**158** (25%)	70	**228**	**–67%**
2029	560	504	**151** (30%)	56	**207**	**–63%**
2030	480	432	**130** (30%)	48	**178**	**–63%**

Illustrative. Treats memory-indexed Spark compute as moving with the server-DRAM contract index; real pass-through is lagged and partly cushioned by long-term agreements, so absolute figures will be lower — the relative gap between the two columns is the point. The 90% routed share is the upside case; a typical estate routes roughly 70%. Every additional 10 points routed moves the blended index roughly 8% closer to the QBricks line. QBricks cost is taken as 20% of the routed memory-indexed spend in 2027, 25% in 2028 and 30% from 2029. Blended = QBricks cost + the unrouted share still on Spark.

Commercial model: gain-share

QBricks is paid from realised savings, not from a licence that grows with your cluster. Compute cost is measured before and after routing on the same workloads, the comparison files are retained for audit, and the agreed share is invoiced against the verified difference.

If the memory curve flattens sooner than modelled, the saving and the fee shrink together. If it does not, you are hedged at a cost that is itself indexed to what you avoided. The hedge is self-funding either way.

What does not change

- The lakehouse remains the system of record and the governance boundary.
- Unity Catalog, Purview or Polaris contracts are enforced on every record — ODCS, human in the loop.
- Large joins, streaming and ML keep running on Spark where it is the right tool.
- Each platform's own security standards apply; agents stay isolated from the data lake.

06 · Conclusion

Treat compute like any other exposure

A treasury would never leave a seven-fold input-price move unhedged for three years. Data platforms have been doing exactly that, because the exposure was hidden inside a billing unit. The memory squeeze is structural, contracted through 2027 and geopolitically fragile beyond it. The fabs will come — but for the window that matters, the cheapest memory is the memory you stop buying.

QBricks lets you keep the lakehouse, keep the governance and drop the cluster for most of the work. Start with a routing assessment on your live workload mix: it produces the actual routed share, the actual blended index and the audit files the gain-share contract will run on.

Request a routing assessment.

Two weeks, on your workload mix, no pipeline build.

qbricks.ai

Sources and notes

1. TrendForce: conventional DRAM contract prices +45–50% QoQ in 4Q25 and +93–98% in 1Q26 (Jun 2026); server DRAM +13–18% forecast for 3Q26, rises continuing into 2H27 (Jul 2026); 4Q26 outlook lifted (Sep 2026). Counterpoint: 64 GB RDIMM $450 to over $900, 4Q25–1Q26. DDR5 chart: 16 Gb chip contract quotes, industry press.
2. DigiTimes via industry press: Samsung, SK hynix and Micron 2027 DRAM and HBM capacity reported fully allocated (Aug 2026).
3. CSIS, "Beyond the Memory Cycle" (Sep 2026): fab lead times of 2–3+ years; HBM takes 3–4× the capacity per bit; Yongin Y1 at 150k wafers/month by end-2028; shortage through 2027–28. TrendForce and IEEE Spectrum: Micron Idaho and Taiwan (2027), Micron Hiroshima and Samsung Pyeongtaek (2028).
4. Hyperscaler capex: company guidance and consensus (Aviva Investors, Vanguard/Bloomberg, Reuters): ~$700–800bn in 2026, above $1tn a year 2027–2030.
5. Bond issuance: Aviva Investors/Bloomberg, ~$170bn hyperscaler issuance to 26 Jun 2026; Reuters/LSEG, ~$121bn for full-year 2025.
6. Chokepoints: ASML sole EUV supplier; Taiwan ~90% of sub-10nm logic; logistics cost uplift 15–22% (Mapshock, Jul 2026); South Korea ~two-thirds of DRAM output (TrendForce); CXMT HBM volume not before 2028 (Counterpoint).
7. AWS Glue pricing (DPU definition and rate); OVHcloud pricing updates (Apr and Sep 2026); The Register on Dell, Lenovo, HP and HPE server repricing (Dec 2025).
8. Memory index scenarios, fab and GW charts: Infinium Consulting modelled estimates, August 2026. Forecasts, not actuals. QBricks routing shares and savings are illustrative and depend on the client's workload mix.

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